

AI Engineering Lab Knowledge Survey — Pre-Test

Form A

Why we are asking

The goal of this lab is that you understand how agentic systems work, not merely that your team ships a working harness. To find out whether the lab actually achieves that, we measure what you know about agentic systems *before* the lab starts and *again after* it ends. Comparing the two tells us where the lab teaches well and where we need to improve it. The same questionnaire structure is used at both points in time.

This is voluntary and does not affect your grade

- Participation is entirely **voluntary**. You may skip any question, or stop at any point.
- Your answers have **no influence whatsoever on your grade**. They are not part of the daily reports, the presentations, the written report, or the oral exam.
- The questionnaire is **pseudonymous** such that only you know the pseudonym: do not write your name, your FIM account, or your matriculation number anywhere on this form.
- The knowledge questions are **not an exam**. In the pre-test especially, most topics have not been taught yet, so a low score is entirely expected. Please do not guess wildly — use the “I do not know” box instead. That is more useful to us than a lucky guess.
- The full answer key will be published after the post-test, so you can learn from them.

Data protection

We process only the answers on this form, linked by the self-generated code below — no names and no identifiers that let us contact or identify you. Data are stored on university infrastructure, evaluated in aggregate, and reported only in aggregate. No result will be reported in a way that could single out an individual (groups smaller than five are not reported separately). Legal basis is your consent, which you may withdraw — though once forms are pooled we can no longer locate an individual questionnaire, so withdrawal is only possible before pooling. Questions about the survey or the data: steffen.herbold@uni-passau.de.

☐ **I have read the information above and agree to take part.**

(If you do not agree, please hand in the form blank so that we can account for it.)

Your personal code

We need to match your pre-test to your post-test, but we do not want to know who you are. Build your code from the four rules below: they give the same result on both occasions, and we cannot reverse them. One character per box, in CAPITAL letters.

Box 1 First letter of your **mother's** first name

Box 2 First letter of your **father's** first name

Boxes 3–4 **Day** of your own birthday, two digits (e.g. the 7th → 07)

Box 5 Last digit of the **house number** you grew up in

Your code:

Example: mother Anna, father Bernd, born on the 23rd, house number 14 → AB234

Part A Demographic Data

These questions characterise the group as a whole and let us account for differences in starting points.

A1 Degree programme currently enrolled in:

- ☐ M.Sc. Computer Science ☐ M.Sc. Artificial Intelligence Engineering
☐ Other: _____

A2 Semester you are currently in: ☐ 1–2 ☐ 3–4 ☐ 5–6 ☐ 7–8 ☐ 9 or more

A3 Which of these courses have you already passed? *(tick all that apply)*

- ☐ Foundations of AI Engineering
☐ Introduction to Deep Learning
☐ Deep Learning for Natural Language and Code
☐ Principles of AI Engineering
☐ Lab Courses on Machine Learning and Data Science
☐ Additional Lectures on Machine Learning and Data Science

☐ Software Engineering
☐ Software Testing
☐ Lab Course of Software Engineering with AI
☐ Additional Lectures on Software Engineering

☐ None of the above

A4 Professional software development experience (internship, working student, employment — excluding coursework!): ☐ none ☐ <1 year ☐ 1–2 years ☐ 2–5 years ☐ >5 years

A5 Technical Skills

How would you rate your own ability with each of the following, as of today?

Skill	1	2	3	4	5
Python (beyond scripting, including aspects like modules, packaging, typing)	☐	☐	☐	☐	☐
Git and collaborative version control	☐	☐	☐	☐	☐
Docker or Podman containers	☐	☐	☐	☐	☐
Automated testing with <code>pytest</code>	☐	☐	☐	☐	☐
Consuming and designing REST/HTTP APIs	☐	☐	☐	☐	☐
Asynchronous or concurrent programming	☐	☐	☐	☐	☐
Metrics and monitoring (e.g. Prometheus, Grafana)	☐	☐	☐	☐	☐
Reading and writing technical documentation in English	☐	☐	☐	☐	☐

1 = no experience 2 = beginner 3 = competent 4 = proficient 5 = expert

A6 Experience with LLMs and AI tools

How would you rate your current exposure to LLMs and AIs?

How often do you currently use ...	daily	weekly	monthly	tried once	never
a chat interface to an LLM (e.g. ChatGPT, Claude, Gemini)	☐	☐	☐	☐	☐
an AI coding assistant inside your editor or IDE (e.g., GitHub CoPilot, Cursor)	☐	☐	☐	☐	☐
an agentic coding tool that runs commands on its own (e.g., Claude Code, OpenAI Codex)	☐	☐	☐	☐	☐

A7 Experience with programming software with LLM components*Have you experience with programming software that uses LLMs or other AI components?*

Have you ever, yourself, ...	yes	attempted	no
called an LLM API directly from your own code?	☐	☐	☐
written a prompt that defines a system role or tool schema?	☐	☐	☐
built a retrieval-augmented generation (RAG) pipeline?	☐	☐	☐
implemented a loop in which an LLM calls tools autonomously?	☐	☐	☐
written or extended an MCP server?	☐	☐	☐
instrumented an application with metrics or traces?	☐	☐	☐
run untrusted or model-generated code inside a sandbox?	☐	☐	☐

A8 Where does your current knowledge about LLMs and agents mainly come from?*(tick all that apply)*

- ☐ university courses ☐ self-study (docs, papers, blogs, videos) ☐ work or internship
☐ hobby projects ☐ conversations with peers ☐ I have essentially no prior knowledge

Part B Your own assessment of your knowledge

This part asks what you think you know. Answer as of today.

B1 The learning objectives of this lab

Judge your current knowledge

Today, I would say ...	1	2	3	4	5
I know what the components of an agentic system are, how they work, and the role each one fulfils.	☐	☐	☐	☐	☐
I understand how these components interact with each other, and how those interactions extend what an agent can do.	☐	☐	☐	☐	☐
I understand how to ensure the quality of the individual components of an agentic system, and of the system as a whole.	☐	☐	☐	☐	☐
I understand which telemetry data is relevant for monitoring an agentic system.	☐	☐	☐	☐	☐
I could implement, monitor, and maintain an agentic system, including its source code, tests, and telemetry.	☐	☐	☐	☐	☐

1 = strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = strongly agree

B2 Individual components

*For each component, give two ratings: how well you could **explain** it to a fellow student, and how well you could **implement** it yourself in code without prior research. These are deliberately separate — being able to explain something is not the same as being able to build it.*

Component	I could explain it					I could implement it				
	1	2	3	4	5	1	2	3	4	5
The role of the LLM in an agentic system	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
The agentic loop and its control flow	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Context and context engineering	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Tools and tool interfaces	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
The Model Context Protocol (MCP)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Skills	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Long-term memory	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Hooks	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission management	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Sandboxing	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Sub-agents and agent-to-agent communication	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Observability and telemetry for agents	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Testing and quality assurance of agentic systems	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1 = not at all 2 = a little 3 = partly 4 = mostly 5 = fully

B3 Confidence in specific situations

Consider what you would feel comfortable doing as part of professional work.

I am confident that I could ...	1	2	3	4	5
design the architecture of an agent harness from scratch, given a set of requirements.	☐	☐	☐	☐	☐
work out why a particular agent run failed, given its logs and traces.	☐	☐	☐	☐	☐
decide what should be measured to operate an agentic system in production.	☐	☐	☐	☐	☐
judge whether someone else's agent harness is well built.	☐	☐	☐	☐	☐
explain to a non-expert why an agent needs a harness at all.	☐	☐	☐	☐	☐
1 = strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = strongly agree					

Part C Knowledge questions

20 questions, each with exactly one best answer. Tick one box per question. If you do not know, please tick "I do not know" rather than guessing — an honest blank is more useful to us than a lucky guess, and none of this affects your grade. Plan for about 15 minutes.

- The LLM in an agentic system is usually described as *stateless across calls*. What does that mean?
 - Its weights are frozen, so it cannot be adapted to new data.
 - It retains nothing from one call to the next, so everything it needs must be supplied in the current request.
 - The provider stores the conversation and reloads it automatically for each call.
 - It returns the same output for the same input, because it holds no random state.☐ I do not know.
- Why does a harness try to keep the context small even in runs that are nowhere near filling the context window?
 - Because model performance degrades as the window fills, so the goal is the smallest set of high-signal tokens that still produces the desired behaviour.
 - Because tokens that are not needed in the current step are discarded by the model anyway.
 - Because the context window is shared between all agents using the same endpoint.
 - Because a shorter context guarantees that the loop terminates in fewer iterations.☐ I do not know.
- A run has grown so long that the context no longer fits. The team wants to keep the agent coherent while spending fewer tokens. Which pair of measures matches this goal?
 - Raise the temperature so the model relies less on the context, and shorten the system prompt.
 - Split the context evenly across several LLM calls and merge the answers afterwards.
 - Summarise the older turns, and move state that is not needed right now to external storage that can be read back on demand.
 - Drop the tool schemas from the request, since the model has already used those tools once.☐ I do not know.

4. Why does the loop need guard conditions such as an iteration cap or a token budget?
- (a) Because the LLM may return neither a final answer nor a valid action, and without a bound the loop would never terminate.
 - (b) Because the model provider charges a penalty for requests above a fixed rate.
 - (c) Because every iteration adds a tool to the registry, which would eventually exhaust the schema limit.
 - (d) Because the user must be given a chance to intervene after a fixed number of steps.
- ☐ *I do not know.*
5. A team extends their loop so that the agent first writes a plan for the task and then executes it. What must the loop now do that a plain ReAct loop does not?
- (a) Call the LLM twice per iteration, once for reasoning and once for acting.
 - (b) Persist the plan to long-term memory before any tool may be called.
 - (c) Disable its iteration cap, because the plan already bounds the number of steps.
 - (d) Contrast the progress made so far against that initial plan on each pass through the loop.
- ☐ *I do not know.*
6. Which of the following best describes a *tool* in an agentic system?
- (a) A store of facts that survives from one agent run to the next.
 - (b) An interface with a defined schema through which the agent retrieves information or acts on an environment.
 - (c) A prompt template that instructs the model how to solve a class of tasks.
 - (d) The component that decides which action the agent should take next.
- ☐ *I do not know.*
7. A `read_file` tool returns an empty string both when the file is empty and when the path does not exist. What is the most likely consequence for the agent?
- (a) It retries the call with the same arguments until the loop's iteration cap is reached.
 - (b) The tool registry rejects the call, because an empty result violates the tool's schema.
 - (c) It cannot tell the two situations apart and reasons on from a wrong observation, because the failure was never communicated to it.
 - (d) The loop enters its terminal state, because an empty observation carries no information.
- ☐ *I do not know.*
8. What is the Model Context Protocol (MCP)?
- (a) A language model specialised in managing an agent's context window.
 - (b) A protocol through which an agent connects to servers that expose tools and other capabilities.
 - (c) A file format for storing an agent's long-term memory between runs.
 - (d) A specification of how much context a model must be able to process.
- ☐ *I do not know.*

9. A requirement says the tools of an MCP server must be made available to the model *without hard-coding individual tools*. What does the harness have to do?
- (a) Generate a Python wrapper function per tool before the application is built.
 - (b) Keep a configuration file that lists every tool name the server is known to provide.
 - (c) Ask the model which tools it thinks exist and validate its guesses against the server.
 - (d) Query the server for its tool list at runtime and pass the returned schemas on to the model.
- ☐ I do not know.
10. Why does a harness usually *not* keep all available skills in the context at all times?
- (a) Because skills may contradict one another if more than one is loaded at a time.
 - (b) Because the model can execute only one skill per run.
 - (c) Because a skill has to be validated by a hook before it may enter the context.
 - (d) Because the context should carry only what the current task needs; a skill is disclosed when the task calls for it.
- ☐ I do not know.
11. What characterises long-term memory, as opposed to context?
- (a) It is the part of the context window that is never summarised away.
 - (b) It is the conversation history of the current run, kept in full.
 - (c) It lives outside the context window, persists across runs, and is consulted on demand.
 - (d) It is the larger context window offered by models that support one.
- ☐ I do not know.
12. What is a hook in an agent harness?
- (a) A tool the agent may call to inspect its own state.
 - (b) A custom function that runs at a defined point in the agent's lifecycle, such as before a tool call.
 - (c) A connection to an external server from which tools are discovered.
 - (d) A rule that decides whether the agent needs the user's approval.
- ☐ I do not know.
13. A team wants to block any tool call whose arguments point outside the sandbox root, *before* it is executed. Which mechanism is designed for that?
- (a) A pre-use hook that inspects the arguments and can reject the call.
 - (b) A post-use hook that reports the violation once the tool has run.
 - (c) A sub-agent that reviews the main agent's proposed actions.
 - (d) A longer system prompt instructing the model never to leave the sandbox.
- ☐ I do not know.

14. In a tiered permission model, which operation belongs in the tier that requires mandatory confirmation by the user?
- (a) Listing the files in the agent's working directory.
 - (b) Running the project's unit tests inside the container.
 - (c) Reading a configuration file in the sandbox.
 - (d) Opening a pull request against a public repository.
- ☐ I do not know.
15. A permission policy supports the outcomes *allow*, *deny* and *require user confirmation*. Why is the third outcome needed rather than deciding every case in advance?
- (a) Because the policy engine cannot evaluate its rules fast enough to keep the loop responsive.
 - (b) Because the model cannot produce arguments that a static rule would be able to check.
 - (c) Because for some actions the consequences depend on the situation, so the judgement has to be handed to a human at the moment it arises.
 - (d) Because denying an action would end the run, which the harness has to avoid.
- ☐ I do not know.
16. Besides containment, what second benefit does a stable sandbox provide?
- (a) Lower latency, because the sandboxed environment sits closer to the model endpoint.
 - (b) A guarantee that the agent cannot loop indefinitely.
 - (c) Automatic approval of low-risk actions without asking the user.
 - (d) Reproducibility: the same command, seed and dependency set can be replayed under comparable conditions, so a failure reflects a real defect rather than environment drift.
- ☐ I do not know.
17. A main agent delegates a subtask to a specialised sub-agent. What does the main agent get back, and why does that matter?
- (a) A condensed result, so that the sub-agent's detailed intermediate work never enters the main agent's context.
 - (b) The sub-agent's full transcript, so that the main agent can verify every step it took.
 - (c) The sub-agent's tool schemas, so that the main agent could repeat the work itself.
 - (d) Nothing — the sub-agent writes its result to long-term memory, which the main agent reads later.
- ☐ I do not know.
18. Which set of measurements is most relevant for monitoring an agentic system in operation?
- (a) Run throughput and latency, success and failure rates, model and token usage, tool usage, and permission decisions.
 - (b) CPU temperature, disk fragmentation, and the number of open file descriptors.
 - (c) The number of lines of code in the harness and its test coverage.
 - (d) The model's parameter count, context window size, and training cut-off date.
- ☐ I do not know.

19. Your dashboard shows that the tool failure rate rose from 2% to 20% this morning. What do you need *in addition* to the aggregate metrics in order to find the cause?

- (a) A longer retention period for the same aggregate counters.
- (b) A second dashboard showing the same metrics with a finer colour scale.
- (c) Structured logs or traces that let you inspect what happened inside individual runs.
- (d) More frequent scraping, so that the same counters are sampled at a finer time resolution.

☐ *I do not know.*

20. An end-to-end test instructs the agent to create `output/answer.md` containing the text **forty-two**. Which assertion is robust against the agent's non-determinism?

- (a) The agent called `create_folder` before `create_file`.
- (b) The file exists at that path and its contents are exactly the expected text.
- (c) The agent needed exactly two tool calls to complete the task.
- (d) The agent's final message contains the sentence "I have created the file."

☐ *I do not know.*